

FRIEDRICHSHAFEN, Germany

WMO: 109350

Lat: 47.671N

Lon: 9.511E

Elev: 1367

StdP: 13.98

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.5	19.4	8.3	8.8	18.8	14.1	11.8	21.5	25.3	43.6	22.7	41.2	7.8	30	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	87.7	68.3	84.3	66.6	80.7	65.9	70.2	82.7	68.3	80.0	67.0	78.2	6.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	101.1	76.5	64.4	95.2	73.3	62.8	89.8	71.5	34.9	82.8	33.6	80.5	32.3	77.9	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.0	16.3	13.9	DB	10.9	93.2	4.2	2.1	7.9	94.7	5.5	95.9	3.2	97.1	0.2	98.6
			WB	10.2	73.2	4.3	2.1	7.2	74.7	4.7	75.9	2.3	77.1	-0.8	78.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	33.5	33.9	42.4	50.3	56.7	64.5	67.5	66.4	58.7	50.6	41.9	35.4	(6)
	DBStd	13.70	6.75	7.78	6.56	6.93	6.56	6.37	5.66	5.69	5.22	6.07	6.05	6.36	(7)
	HDD50	2074	511	450	242	81	16	0	0	0	1	69	252	452	(8)
	HDD65	5687	976	870	699	442	265	87	39	53	200	446	693	917	(9)
	CDD50	2166	1	0	8	90	223	434	542	509	262	88	9	0	(10)
	CDD65	305	0	0	0	1	7	72	117	97	11	0	0	0	(11)
	CDH74	2933	0	0	0	24	123	739	1117	843	83	4	0	0	(12)
	CDH80	878	0	0	0	3	16	228	375	252	4	0	0	0	(13)
(14) Wind	WSAvg	5.3	5.7	5.8	6.5	5.5	5.6	5.2	5.2	4.5	4.8	4.8	5.1	5.2	(14)
(15) Precipitation	PrecAvg	39.2	2.2	2.3	2.3	3.1	3.9	4.9	4.6	4.9	3.0	2.5	3.1	2.2	(15)
	PrecMax	48.6	4.4	6.3	4.8	6.4	7.0	8.0	9.4	12.0	9.4	5.2	8.3	5.0	(16)
	PrecMin	30.9	0.7	0.3	0.6	1.0	1.3	2.1	2.0	2.1	0.8	0.5	0.8	0.2	(17)
	PrecStd	4.5	1.0	1.3	1.2	1.3	1.6	1.7	2.0	2.3	2.1	1.5	1.9	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	57.4	66.6	76.9	82.3	90.0	93.1	91.1	80.4	71.6	64.8	55.2	(19)
		MCWB	47.4	46.3	51.0	57.4	62.1	70.2	68.2	67.9	66.0	57.8	51.4	48.5	(20)
	2%	DB	48.5	51.9	62.5	73.0	77.2	86.0	88.0	86.1	76.8	66.6	58.9	50.4	(21)
		MCWB	44.5	43.9	48.9	55.3	60.5	66.5	68.1	68.8	64.9	57.8	51.7	45.6	(22)
	5%	DB	45.0	48.3	58.6	69.5	73.6	82.2	84.3	82.5	73.0	63.0	53.9	46.8	(23)
		MCWB	42.0	42.5	47.3	54.4	58.7	65.3	66.3	67.4	62.9	55.9	49.4	43.4	(24)
	10%	DB	42.8	44.7	54.0	64.5	70.0	77.4	80.5	78.8	69.4	60.5	51.4	44.6	(25)
		MCWB	40.4	40.0	45.4	52.2	57.7	63.5	65.6	66.5	61.2	55.2	47.6	42.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.5	48.7	52.7	59.0	65.3	73.0	71.1	72.5	67.7	60.5	53.7	48.8	(27)
		MCDB	52.0	54.9	63.0	73.0	78.2	87.3	85.3	84.2	77.2	68.6	60.7	52.6	(28)
	2%	WB	44.8	45.0	49.8	56.6	62.6	69.0	69.7	70.3	65.5	58.5	51.3	46.4	(29)
		MCDB	47.7	50.8	60.3	68.8	73.2	81.7	83.5	81.6	74.5	65.0	57.4	50.0	(30)
	5%	WB	42.2	42.6	48.2	54.5	60.2	66.7	67.9	68.6	63.4	57.0	49.9	44.1	(31)
		MCDB	44.5	47.4	56.4	67.0	71.1	78.8	80.6	79.2	71.3	62.4	54.6	46.9	(32)
	10%	WB	40.0	40.2	46.2	52.8	58.1	64.7	66.4	67.0	61.6	55.4	47.9	41.7	(33)
		MCDB	42.1	44.1	53.3	63.9	67.8	76.2	78.5	77.0	68.1	60.0	51.1	43.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	12.6	17.4	19.9	18.3	18.8	19.5	19.5	17.3	15.2	10.9	10.1	(35)
		MCDBR	14.1	20.7	26.2	28.1	26.1	27.2	27.7	26.3	22.7	20.4	17.2	14.2	(36)
		MCWBR	10.8	13.6	14.5	14.2	12.2	11.8	10.8	11.4	11.9	12.5	11.2	10.4	(37)
	5% WB	MCDBR	13.2	19.1	23.7	26.6	23.1	24.6	24.8	23.6	21.2	18.8	16.1	13.1	(38)
		MCWBR	10.5	13.0	13.5	13.8	11.2	11.4	10.5	11.3	11.7	12.3	11.0	10.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.292	0.317	0.354	0.392	0.405	0.411	0.413	0.406	0.378	0.348	0.316	0.288	(40)
	taud		2.430	2.361	2.337	2.266	2.274	2.296	2.286	2.323	2.378	2.452	2.484	2.443	(41)
	Ebn at Noon		252	267	273	272	272	271	268	265	262	253	240	239	(42)
	Edh at Noon		21	28	34	40	41	40	40	37	32	25	20	18	(43)
(44) All-Sky Solar Radiation	RadAvg		351	613	995	1367	1585	1816	1744	1524	1138	710	390	287	(44)
	RadStd		39	86	134	194	161	166	159	156	118	76	47	45	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (110 neighbors)	N/A	N/A	N/A	N/A	+0.77	+0.67	N/A	N/A	+97	+25

Nomenclature: See separate page